**Objectives:** In this study, we aimed to integrate spatially-resolved CT-extracted radiomic profiles with matched histopathological and molecular data to decipher the cellular programs underlying the spatial heterogeneity in SSc-ILD, and to assess how these cellular programs are reflected in radiomic phenotypes.
**Methods:** The whole right lung of a 75-year-old female SSc-ILD patient with shortness of breath, dry cough, compromised lung function and fibrotic lesions on CT was systematically dissected into 65 well-defined samples within 5 hours post-mortem (Figure 1).
The same sampling scheme was applied to segment the clinical pre-operative HRCT scan and 1'386 radiomic features were extracted for every 3D sample volume using the open-source software Z-rad. Spatial association of regional radiomics profiles was performed with CT-colocalized histopathology, including fibrosis extent, evaluated using Ashcroft scoring and morphological patterns, and molecular data from single-nuclei RNA-sequencing (snRNAseq). Single-cell data were analysed using "scanpy" following current single-cell best practices. Differential cellular abundance testing, pathway activation and cell-cell communication were inferred using "Milo", "GSEApy" and "CellChat", respectively.
**Results:** Within the lung, radiomic profiles significantly differed with spatial location, revealing a typical apical to basal gradient characteristic for the spatial heterogeneity in SSc-ILD as evidenced by principal component analysis and unsupervised clustering. To assess whether those differences in radiomic profiles would also translate to the molecular level, including changes in cellular composition, we performed snRNAseq of 14 distal lung regions, representatively sampled from the lung's apex to base. We profiled more than 124,000 cells and identified more than 50 different cell types and states. We observed striking location-specific alterations in cellular composition, gene expression profiles and cellular crosstalk/cell-cell communication networks, particularly in the alveolar niche. Spatial location-resolved relative cell frequency analysis revealed significant changes in cell type composition consistent with the corresponding radiomic and histopathological phenotype. For instance, we found higher frequencies of immune cell types, such as T and dendritic cells in highly inflammatory regions and myofibroblasts in fibrotic regions. Integration of radiomics and transcriptomics data revealed spatial radiomic modules with distinct cellular profiles, including an apical-predominant T cell exhaustion and a basal-predominant macrophage-driven fibrogenic program.
**Conclusion:** Our unique approach for in-depth tissue phenotyping provides novel pathophysiological insights into the spatial heterogeneity of SSc-ILD. Furthermore, our study underscores the potential of radiomics as virtual organ-scale biopsies, offering a comprehensive understanding of disease patterns at both macroscopic and molecular levels.
**REFERENCES: NIL.**

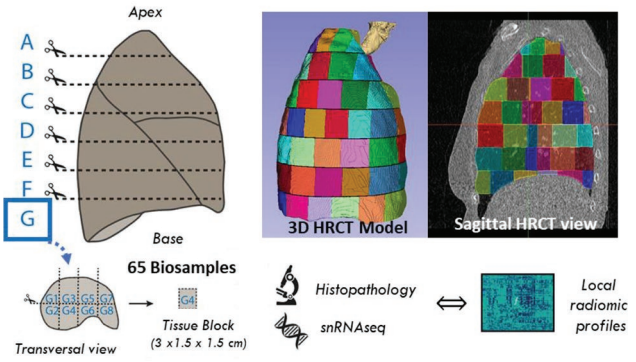


**Figure 1.** Systematic dissection of the whole right lung of a SSc-ILD patient into 65 well-defined samples for histopathological and single nuclei transcriptomic profiling. The same sampling strategy for segmentation was applied to the clinical HRCT scan. This enables the spatial association of the local radiomics profiles and CT-colocalized histopathological and molecular data.

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## AB1146 ALTERNATIVE SPLICING IDENTIFICATION IN SYSTEMIC SCLEROSIS AND THEIR ROLES IN SKIN FIBROSIS

**Keywords:** '-omics, Skin, Fibroblasts

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**Background:** Systemic sclerosis (SSc) is a class of heterogeneous autoimmune diseases, and fibrosis is at the centre of the pathogenic triad of SSc. Alternative splicing (AS) is a post-transcriptional regulatory process in which specific exons and/or introns of precursor messenger RNA (pre-mRNA) are selectively included or excluded to generate multiple mature mRNA isoforms. AS encompasses several modes, including splicing exons (SE), alternative 3' splicing site (A3SS), alternative 5' splicing site (A5SS), intron retention (IR) and mutually exclusive exons (MXE). AS and IR have recently been recognized as playing important roles in many disorders., but little is known about them in systemic sclerosis and fibrotic disorders.
**Objectives:** To identify AS and IR profiles in SSc, assess their impact on protein and explore their roles in the process of skin fibrosis.
**Methods:** A comprehensive genome-wide analysis of AS and IR was conducted across SSc skin tissue and fibroblasts based on bulk RNA-seq, single-cell RNA-seq, and 4D label-free mass spectrometry. AS events were identified using software rMATS. IR expression was analyzed using iread and validated by real-time PCR (RT–PCR). Correlations among clinical features, introns, splicing factors (regulators of AS) and proteins were assessed by spearman correlation or the LASSO method.
**Results:** Significant changes were noted in AS profiles within SSc skin tissue, with a particular emphasis on the noteworthy alterations in IR. The levels of AS events and IR demonstrated correlations with the extracellular matrix score, SSc autoantibodies, disease duration, and the clinical local skin score. Beyond skin tissues, distinct IR profiles were discernible among different skin fibroblast subtypes. Upon TGF-β stimulation, fibroblasts experienced substantial changes in IR profiles, influencing genes associated with fibrosis-related pathways, notably collagen fibril organization. An overarching integrated analysis involving introns, exons, and proteome profiles indicated that IR mainly exerted a negative impact on protein expression, with certain changes being intronically regulated. However, IR also manifested other effects on protein expression. Additionally, the regulatory network of splicing factors displayed distinct patterns in both SSc skin tissue and fibroblast subpopulations.
**Conclusion:** Altered AS and IR profiles, along with dysfunctional splicing machinery, were observed in skin fibroblasts of SSc. These findings suggest the potential to directly impact protein expression and may be associated with the fibrosis process. This insight could contribute to a better understanding of the pathogenesis of SSc.
**REFERENCES: NIL.**
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## AB1147 DIFFERENTIAL RECOGNITION OF TOPOISOMERASE 1 (TOP1)-DNA COMPLEXES MODULATES ACTIVATION OF TOP1-REACTIVE B CELLS IN SYSTEMIC SCLEROSIS

**Keywords:** Autoantibodies, Biomarkers, Adaptive immunity

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**Background:** Systemic sclerosis (SSc) is hallmarked by autoreactive B cell responses against nuclear antigens. To date, it is poorly understood if and how these B cell responses contribute to disease. A subset of SSc patients harbors B cells targeting the nuclear enzyme topoisomerase 1 (TOP1). Anti-TOP1 autoantibodies (ATAs) associate with severe disease marked by progressive lung and skin fibrosis. Patients with lung fibrosis harbor TOP1-reactive B cells with an activated phenotype [1]. Here, we hypothesized that mechanisms underlying the activation of TOP1-reactive B cells could determine the development of disease and drive its progression. Importantly, DNA binding by TOP1 generates

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## AB0209 A STUDY OF CORRELATION BETWEEN PLATELETS AND LYMPHOCYTE SUBSETS IN SCLEROSIS

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**Background:** Systemic sclerosis (SSc) is an autoimmune disease characterized by vasculopathy, autoimmunity and widespread dermal and visceral fibrosis. There are existing evidence to support excessive platelet activation and their contribution to vascular function and fibrosis. Now we will focus on the immune role of PLT in SSC.
**Objectives:** By analyzing the correlation between platelet and lymphocyte subsets, CD4 + T cell subsets and disease activity in patients with diffuse SSc (dcSSc)/limitedSSc (lcSSc), and to explore the immune role of PLT in SSC.
**Methods:** The peripheral blood of 21 stable disease patients, 31 active disease patients and 20 healthy controls (HC) were collected. The clinical data and laboratory indicators of them were enrolled. The T, B, NK lymphocyte subsets and CD4 + T cell subsets were detected by flow cytometry (FCM). The CD4 + T cell subsets contains Th1, Th2, TH17, Treg, Th1/Th2 and Th17/Treg. Non-parametric Kruskal-Wallis H test was performed on multiple independent samples. The correlation between variables was used by Spearman correlation analysis.
**Results:** The PLT,PCT,MPV,PDW in peripheral blood of the stable group and active group were significantly higher than the HC group($P<0.05$). The amount of T cells in peripheral blood of active group were [1127.80 (796.66,1363.79)], which was lower than the percentage of HC group [1246.44(984.81,1497.84)] and stable group [1428.59(1179.09,2022.88)], there was significant difference between active group and stable group ($Z=6.525$,$P=0.038$). The amount of NK cells of the stable group [198.48 (140.22, 302.97)]and active group [143.76(89.00, 230.00)]were significantly lower than the HC group [224.81(155.04,350.51)]($Z=8.694$, $P<0.05$). The amount of Th cells of the stable group[888.11(679.55, 1430.70)] and active group[684.20(385,878)] were higher than the HC group[574.84 (493.22,728.00)], the former was significant, there was significant difference between active group and stable group($Z=8.694$, $P<0.05$). The Th17/Treg ratio of the stable group [0.18 (0.10, 0.34)] and active group [0.21(0.13, 028] were lower than the HC group [0.25 (0.12,0.31)], there was significant difference between active group and HC group($Z=0.508$, $P<0.05$). The ESR and CRP were positively correlated with PLT, PCT; The amount of Th cells and Th17 cells were positively correlated with PLT,PCT; The amount of Th2 cells and the ratio of CD4+T/CD8+ T were positively correlated with PCT; There was a negative correlation between the ratio of Th1/Th2 and PCT; The amount of Th2 cells was positively correlated with MVP, and the ratio of Th1/Th2 was negatively correlated with MPV.Among these immune cells, NK cells was negatively correlated with ESR and DSS(disease severity scale).
**Conclusion:** In scleroderma, NK cells are associated with the severity of the disease, and PLT affects the number of Th1/Th2, Th17, and disease activity.

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## AB0210 THE GLOBAL EXPRESSION OF MIRNAS AND LNCRNAS IN THE EXOSOMES OF SYSTEMIC SCLEROSIS PLASMA AND NEUTROPHIL AND RELATED FUNCTIONS

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**Background:** Systemic sclerosis (SSc) is a systemic autoimmune disease with unknown pathogenesis. Exosomes (EXOs) are cell-derived vesicles 30-150 nm in size that contain various mRNAs, microRNAs (miRNAs), long non-coding RNAs (lncRNAs) and proteins. Plasma EXOs play wide roles in various diseased, however, little is known in SSc.
**Objectives:** Investigate the global expression of miRNAs and lncRNAs in the EXOs of SSc plasma and neutrophil. Explore the potential function of EXOs in the pathogenesis of SSc.
**Methods:** EXOs were respectively isolated from plasma, cultured neutrophil supernatants, and were identified by transmission electron microscopy. Global expression of miRNAs and lncRNAs form 5 SSc and 5 normal controls was analysis by IlluminaHiSeq 3000 platform. Differentiated expressed and bioinformatics analysis was performed by edgeR, limma package, GO, KEGG, and cytoscape. Furthermore, we used the EXOs stimulated human dermal microvascular endothelial cells (HDMECs) and human primary skin fibroblasts, and explore the potential functions of EXOs.
**Results:** 1. In plasma EXOs, we identified a total of 37 miRNAs and 479 lncRNAs that showed significant differences between the two groups. Among them, 26 of upregulated miRNAs were involved in the ErbB signalling pathway; 11 of downregulated miRNAs were involved in the MAPK signalling pathway, TGF-β signalling pathway, AMPK signalling pathway;299 of upregulated lncRNAs were involved in the PI3K-Akt signalling pathway, p53 signalling pathway;180 of downregulated lncRNAs were involved in lysosome.
2. In neutrophil EXOs, we identified a total of 22 miRNAs and 281 lncRNAs that showed significant differences between the two groups. Among them, 12 of upregulated miRNAs were involved in the Wnt signalling pathway; 10 of downregulated miRNAs were involved in the MAPK signalling pathway, AMPK signalling pathway; 119 of upregulated lncRNAs were involved in interleukin-23 signalling pathway;162 of downregulated lncRNAs were involved in the signalling by GPCR, signalling by NOTCH and TRAIL.
3. After stimulated with SSc plasma EXOs, the expression levels of has-miR-324-5p and SP1/SMAD2, has-miR-624-5p and PIK3RCB, has-miR-483-5p and MAPK1 were negatively correlated, the expression levels of ENST00000562409.1 and OSMR, ENST00000601511.1 and HMGB1/NRAS were positively correlated, variably in HDMECs and human primary skin fibroblasts.
4. After stimulated with SSc neutrophil EXOs, the expression levels of has-miR-1268a and PRKCA, has-miR-299-3p and IGF1R were negatively correlated, the expression levels of ENST00000520562.1 and IL23R, ENST00000596567.1 and TFDP2, ENST00000608572.1 and HDAC2 were positively correlated, variably in neutrophil, HDMECs and human primary skin fibroblasts.
5. These miRNAs and lncRNAs with consistent expression might be involved in the pathogenesis of SSc EXOs.
**Conclusion:** Our study identified and confirmed differentially miRNAs and lncRNAs in the neutrophil EXOs and plasma EXOs. Those genes may be involved in the pathological mechanism of SSc, such as has-miR-324-5p and SP1/SMAD2, has-miR-624-5p and PIK3RCB, has-miR-483-5p and MAPK1, ENST00000520562.1 and IL23R, ENST00000596567.1 and TFDP2, ENST00000608572.1 and HDAC2.
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## AB0211 NEUTROPHIL-DERIVED EXOSOME S100A8/A9 INHIBITS ANGIOGENESIS IN SYSTEMIC SCLEROSIS

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**Background:** Systemic sclerosis (SSc) is an autoimmune disease characterized by vasculopathy, inflammation, and extensive fibrosis of the skin and organs. Exosomes (EXOs) are cell-derived vesicles 30-150 nm in size that contain various mRNAs, microRNAs and proteins.
**Objectives:** Here, we aimed to investigate the roles of EXOs in SSc pathogenesis, especially in angiogenesis.
**Methods:** EXOs were respectively isolated from plasma, cultured peripheral blood mononuclear cells (PBMCs) and neutrophil supernatants, and were identified by transmission electron microscopy. The expression of S100A8/A9 was measured by real-time PCR and ELISA. Proliferation, migration and scratch assays in human dermal microvascular endothelial cells (HDMECs) were used to study the influence of neutrophil EXOs and neutrophil EXOs S100A8/A9. We also performed a genome-wide transcriptome analysis on PBMCs from 19 SSc patients and 18 matched normal controls (NC) using Illumina BeadChip arrays. The ingenuity pathway analysis (IPA) tool and Database for Annotation, Visualization and Integrated Discovery (DAVID) were used for bioinformatics analysis.
**Results:** Plasma EXOs and neutrophil EXOs from SSc patients suppressed the proliferation and migration of HDMECs. Using a microarray analysis we found 28% genes upregulated in PBMCs could exist in EXOs, especially the S100 protein family, including S100A8/A9. High levels of S100A8/A9 were consistently verified in SSc plasma, PBMCs, plasma EXOs, PBMC EXOs and neutrophil EXOs. Particularly, S100A8/A9 expression in neutrophil EXOs was distinctly higher than that in PBMC EXOs in SSc patients. Furthermore, we found that neutrophil EXOs S100A8/A9 inhibit the proliferation and migration of HDMECs, and that might through Toll-like receptor 4 (TLR4) pathway.
**Conclusion:** S100A8/A9 is one of components of neutrophil EXOs that regulates vascular endothelial cell angiogenesis in SSc patients, most likely by activating the TLR4 signalling pathway.
**Acknowledgement:** This study was funded by grants from the National Key Research and Development Program of China (2016YFC0903900), National Natural Science Foundation of China (81671622), Hunan Provincial Natural Science Foundation (2018JJ3823), Clinical Research Fund of Xiangya Hospital Central South University (2014L10) and Independent Innovation Projects of Central South University (2018zzst290).
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## AB0212 LONG-TERM TREATMENT WITH RITUXIMAB IN SYSTEMIC SCLEROSISMANAGEMENT: AN OVERVIEW OF THE CLINICAL EXPERIENCE FROM A REAL-LIFE SETTING

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**Background:** The treatment of systemic sclerosis (SSc) is a clinical challenge because of its complex pathogenesis. The rationale for the use of rituximab (RTX), able to downregulate the B-cell over expression, is demonstrated in different autoimmune diseases, some reports have suggested its key role in regulating both vascular, fibrotic, and immune T- and B-lymphocyte-mediated alterations that characterize several SSc manifestations. According to our previous experience with RTX [1], we suggested a possible therapeutical role of RTX in SSc, along with its good safety profile.
**Objectives:** To investigate the role and efficacy of RTX in our SSc patients' series.
**Methods:** A series of 24 SSc patients (M/F 7/17, mean age 55.2 ±14.7SD years, mean disease duration 11.7± 7.3SD years, L/D cutaneous subsets 11/13) were treated with one or more cycles of RTX (4 weekly infusions of 375 mg/m$^2$) and evaluated during a mean follow-up period of 12.3 ± 6.6SD years, range 2-26 years. The indications to RTX were interstitial lung disease, cutaneous, articular manifestations, evaluating its effects after 6 months of the first cycle and at the end of follow-up.
**Results:** After the first 6 months the extent of skin sclerosis measured with modified Rodnan skin score (mRSS) significantly improved (from 16.3±9.8 to 12.3±7.1; p<0.0001), and remained stable at the end of the follow-up (11.0±6.9; p=0.105). The effect on skin sclerosis was more evident in patients with diffuse cutaneous SSc (n=13) showing a significant decrease of mRSS after the first 6 months (from 24.3± 5.4 to 17.9±4.3; p=0.006) and at the end of the follow-up period (17.8±5.7; p=0.005). Similarly, a valuable improvement of other skin manifestations, namely hypermelanosis (17/20 pts), pruritus (18/21 pts), digital ulcers (8/18 pts) and calcinosis (6/3 pts), was observed. Moreover, arthritis revealed a good response to RTX treatment leading to a clear-cut reduction of swollen (from 2.0±2.8 to 0.3±1.3; p<0.0001) and tender joints (from 5.2±4.0 to 0.9±2.5; p<0.0001) in 20/24 patients. Finally, lung fibrosis detected in 19/24 on chest-CT remained stable during the entire follow-up, as well as pulmonary function tests (PFTs). These positive clinical changes were mirrored by the subjective improvement of patients' well being in all subjects (HAQ from 1.04±0.55 to 0.85±0.38; VAS from 71.0±15.2 to 28.0 ±10.3, p<0.0001). No significant side effects were observed during the entire follow-up. Only in one patient a severe urinary tract infection leading to the discontinuation of the treatment was detected.
**Conclusion:** The present study reinforces the previous trials and our preliminary researches on this topic, showing the efficacy of RTX in the management of SSc with good safety profile. The specific therapeutical role of RTX on B-cell-driven autoimmunity, might explain its beneficial effects on some SSc clinical alterations. The improvement of skin sclerosis, articular symptoms and the stabilization of lung involvement were identified as the main results. Further exploration of the potential clinical efficacy of RTX in SSc with multicentre, double blind, controlled study is needed.

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**DOI:** 10.1136/annrheumdis-2019-eular.5528

## AB0213 RELATED FACTORS TO RENAL INVOLVEMENT IN SYSTEMIC SCLEROSIS PERUVIAN PATIENTS

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**Background:** Systemic sclerosis (SSc) can affect multiple organ systems including the kidney. Renal disease, especially chronic kidney disease (CKD), remains an important cause of morbidity and mortality in SSc. The spectrum of renal complications in SSc include scleroderma renal crisis (SRC), normotensive renal crisis, antineutrophil cytoplasmic antibody associated glomerulonephritis, penicillamine-associated renal disease, and reduced renal functional reserves. Furthermore, subclinical renal impairment affects approximately 10-55% of SSc patients and might be associated with other vascular manifestations. However, the available evidence on CKD in patients with SSc residing in low-middle income countries (LMIC) is scarce. Because the health system of LMIC, and especially Peru, could have great differences in access to diagnosis and management of SSc, it is important to identify which clinical factors would be associated with CKD in patients with this autoimmune disease.
**Objectives:** To identify the associated factors to renal involvement in Peruvian patients with SSc.
**Methods:** We analyzed the associated factors to renal involvement in SSc patients at Hospital Nacional Edgardo Rebagliati Martins Lima-Peru, a national reference hospital in Peru. Between June 2001 and December 2018, we included ambulatory patients, older than 18-year-old with SSc that met the ACR-EULAR classification criteria. In patients who accepted to get informed consent, a complete clinical assessment and a sociodemographic survey were done. Additional clinical data were collected from their clinical records. Multiple Poisson regression with robust standard errors was used to identify significant and independently associated factor. Two models were estimated: one based on theoretically selected variables and another parsimonious model based on variables that was selected using a manual backward selection procedure with a predetermined alpha of 0.2. Adjusted prevalence ratio (aPR), 95% confidence interval and p-values were reported.
**Results:** One hundred and five patients with SSc were included in this study, 15.1% had chronic kidney disease. The average age was 57.8

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